(**c**) Based on your work from part (b), what should δ be set equal to?

(**d**) Write an epsilon-delta proof that $\lim\limits_{z\to 2i}\dfrac{2z^2-3iz+2}{z-2i}=5i$.

53. (**a**) Is it true that $\lim\limits_{z\to z_0}\overline{f(z)}=\lim\limits_{z\to z_0}f(\bar{z})$ for any complex function f? If so, then give a brief justification; if not, then find a counterexample.

(**b**) If $f(z)$ is a continuous function at z_0, then is it true that $\overline{f(z)}$ is continuous at z_0?

54. If f is a function for which $\lim\limits_{x\to 0}f(x+i0)=0$ and $\lim\limits_{y\to 0}f(0+iy)=0$, then can you conclude that $\lim\limits_{z\to 0}f(z)=0$? Explain.

55. (**a**) Prove that the function $f(z)=\text{Arg}(z)$ is discontinuous at every point on the negative real axis.

(**b**) Prove that the function f_1 defined by

$$f_1(z)=\theta,\ -\pi<\theta<\pi$$

is a branch of the multiple-valued function $F(z)=\arg(z)$. [*Hint*: See Example 8.]

56. Consider the multiple-valued function $F(z)=z^{1/3}$ that assigns to z the set of three cube roots of z. Explicitly define three distinct branches f_1, f_2, and f_3 of F, all of which have the nonnegative real axis as a branch cut.

57. Consider the multiple-valued function $F(z)=(z-1+i)^{1/2}$.

(**a**) What is the branch point of F? Explain.

(**b**) Explicitly define two distinct branches of f_1 and f_2 of F. In each case, state the branch cut.

58. Consider the multiple-valued function $F(z)=\left(z^2+1\right)^{1/2}$. What are the branch points (there are two of them) of F? Explain.

Computer Lab Assignments

Reread part (*ii*) of the *Remarks* at the end of Section 3.1. In Problems 59–62, use a CAS to show that the given function is not continuous inside the unit circle by plotting the image of the given continuous parametric curve. (Be careful, *Mathematica* and *Maple* plots can sometimes be misleading.)

59. $f(z)=z+\text{Arg}(z),\ z(t)=-\frac{1}{2}+\frac{1}{2}\sqrt{3}it,\ -1\le t\le 1$

60. $f(z)=\sqrt[4]{r}e^{i\theta/4},\ \theta=\text{Arg}(z),\ z(t)=-\frac{1}{2}+\frac{1}{2}\sqrt{3}it,\ -1\le t\le 1$

61. $f(z)=\sqrt{r}e^{i\theta/2},\ \theta=\text{Arg}(z),\ z(t)=-\frac{1}{2}+\frac{1}{4}e^{it},\ 0\le t\le 2\pi$

62. $f(z)=|z-1|\text{Arg}(-z)+i\,\text{Arg}(iz),\ z(t)=\frac{1}{2}e^{it},\ 0\le t\le 2\pi$

3.2 Differentiability and Analyticity

INTRODUCTION The calculus of complex functions deals with the usual concepts of derivatives and integrals of these functions. In this section we shall give the limit definition of the derivative of a complex function $f(z)$. Although many of the concepts in this section will seem familiar to you, such as the product, quotient, and chain rules of differentiation, there are important differences between this material and the calculus of real functions $f(x)$. As the subsequent chapters of this text unfold, you will see that except for familiarity of names and definitions, there is little similarity between the *interpretations* of quantities such as $f'(x)$ and $f'(z)$.

The Derivative

Suppose $z = x + iy$ and $z_0 = x_0 + iy_0$; then the change in z_0 is the difference $\Delta z = z - z_0$ or $\Delta z = x - x_0 + i(y - y_0) = \Delta x + i\Delta y$. If a complex function $w = f(z)$ is defined at z and z_0, then the corresponding change in the function is the difference $\Delta w = f(z_0 + \Delta z) - f(z_0)$. The **derivative** of the function f is defined in terms of a limit of the difference quotient $\Delta w/\Delta z$ as $\Delta z \to 0$.

DEFINITION 3.2.1 Derivative of a Complex Function

Suppose the complex function f is defined in a neighborhood of a point z_0. The **derivative** of f at z_0, denoted by $f'(z_0)$, is

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \tag{1}$$

provided this limit exists.

If the limit in (1) exists, then the function f is said to be **differentiable** at z_0. Two other symbols denoting the derivative of $w = f(z)$ are w' and dw/dz. If the latter notation is used, then the value of a derivative at a specified point z_0 is written $\left.\dfrac{dw}{dz}\right|_{z=z_0}$.

EXAMPLE 1 Using Definition 3.2.1

Use Definition 3.2.1 to find the derivative of $f(z) = z^2 - 5z$.

Solution Because we are going to compute the derivative of f at any point, we replace z_0 in (1) by the symbol z. First,

$$f(z + \Delta z) = (z + \Delta z)^2 - 5(z + \Delta z) = z^2 + 2z\Delta z + (\Delta z)^2 - 5z - 5\Delta z.$$

Second,

$$\begin{aligned} f(z + \Delta z) - f(z) &= z^2 + 2z\Delta z + (\Delta z)^2 - 5z - 5\Delta z - (z^2 - 5z) \\ &= 2z\Delta z + (\Delta z)^2 - 5\Delta z. \end{aligned}$$

Then, finally, (1) gives

$$\begin{aligned} f'(z) &= \lim_{\Delta z \to 0} \frac{2z\Delta z + (\Delta z)^2 - 5\Delta z}{\Delta z} \\ &= \lim_{\Delta z \to 0} \frac{\Delta z(2z + \Delta z - 5)}{\Delta z} \\ &= \lim_{\Delta z \to 0} (2z + \Delta z - 5). \end{aligned}$$

The limit is $f'(z) = 2z - 5$. ☰

Rules of Differentiation

The familiar rules of differentiation in the calculus of real variables carry over to the calculus of complex variables. If f and g are differentiable at a point z, and c is a complex constant, then (1) can be used to show:

Differentiation Rules

Constant Rules: $\dfrac{d}{dz}c = 0$ and $\dfrac{d}{dz}cf(z) = cf'(z)$ (2)

Sum Rule: $\dfrac{d}{dz}[f(z) \pm g(z)] = f'(z) \pm g'(z)$ (3)

Product Rule: $\dfrac{d}{dz}[f(z)g(z)] = f(z)g'(z) + f'(z)g(z)$ (4)

Quotient Rule: $\dfrac{d}{dz}\left[\dfrac{f(z)}{g(z)}\right] = \dfrac{g(z)f'(z) - f(z)g'(z)}{[g(z)]^2}$ (5)

Chain Rule: $\dfrac{d}{dz}f(g(z)) = f'(g(z))\,g'(z).$ (6)

The **power rule** for differentiation of powers of z is also valid:

$$\frac{d}{dz}z^n = nz^{n-1}, \quad n \text{ an integer.} \tag{7}$$

Combining (7) with (6) gives the **power rule for functions**:

$$\frac{d}{dz}[g(z)]^n = n[g(z)]^{n-1}g'(z), \quad n \text{ an integer.} \tag{8}$$

EXAMPLE 2 Using the Rules of Differentiation

Differentiate:

(a) $f(z) = 3z^4 - 5z^3 + 2z$ **(b)** $f(z) = \dfrac{z^2}{4z+1}$ **(c)** $f(z) = \left(iz^2 + 3z\right)^5$

Solution **(a)** Using the power rule (7), the sum rule (3), along with (2), we obtain

$$f'(z) = 3 \cdot 4z^3 - 5 \cdot 3z^2 + 2 \cdot 1 = 12z^3 - 15z^2 + 2.$$

(b) From the quotient rule (5),

$$f'(z) = \frac{(4z+1)\cdot 2z - z^2 \cdot 4}{(4z+1)^2} = \frac{4z^2 + 2z}{(4z+1)^2}.$$

(c) In the power rule for functions (8) we identify $n = 5$, $g(z) = iz^2 + 3z$, and $g'(z) = 2iz + 3$, so that

$$f'(z) = 5(iz^2 + 3z)^4(2iz + 3).$$

For a complex function f to be differentiable at a point z_0, we know from the preceding section that the limit $\lim\limits_{\Delta z \to 0} \dfrac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$ must exist and equal the same complex number from any direction; that is, the limit must exist regardless of how Δz approaches 0. This means that in complex analysis, the requirement of differentiability of a function $f(z)$ at a point z_0 is a far greater demand than in real calculus of functions $f(x)$ where we can approach a real number x_0 on the number line from only two directions. If a complex function is made up by specifying its real

and imaginary parts u and v, such as $f(z) = x + 4iy$, there is a good chance that it is not differentiable.

EXAMPLE 3 A Function That Is Nowhere Differentiable

Show that the function $f(z) = x + 4iy$ is not differentiable at any point z.

Solution Let z be any point in the complex plane. With $\Delta z = \Delta x + i\,\Delta y$,

$$f(z + \Delta z) - f(z) = (x + \Delta x) + 4i(y + \Delta y) - x - 4iy = \Delta x + 4i\,\Delta y$$

and so
$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{\Delta x + 4i\,\Delta y}{\Delta x + i\,\Delta y}. \tag{9}$$

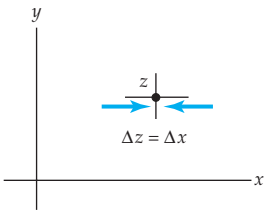
Now, as shown in FIGURE 3.2.1(a), if we let $\Delta z \to 0$ along a line parallel to the x-axis, then $\Delta y = 0$ and $\Delta z = \Delta x$. Along this path we have:

$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{\Delta x}{\Delta x} = 1. \tag{10}$$

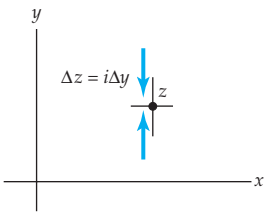
On the other hand, if we let $\Delta z \to 0$ along a line parallel to the y-axis as shown in Figure 3.2.1(b), then $\Delta x = 0$ and $\Delta z = i\,\Delta y$ so that

$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{4i\,\Delta y}{i\,\Delta y} = 4. \tag{11}$$

In view of the obvious fact that the values in (10) and (11) are different, we conclude that the limit in (9) does not exist. Therefore, $f(z) = x + 4iy$ is nowhere differentiable; that is, f is not differentiable at any point z. ☰



(a) $\Delta z \to 0$ along a line parallel to x-axis



(b) $\Delta z \to 0$ along a line parallel to y-axis

FIGURE 3.2.1 Approaching z along a horizontal line and then along a vertical line

The basic power rule (7) does not apply to powers of the conjugate of z because, like the function in Example 3, the function $f(z) = \bar{z}$ is nowhere differentiable. See Problem 21 in Exercises 3.2.

Analytic Functions Even though the requirement of differentiability is a stringent demand, there is a class of functions that is of great importance whose members satisfy even more severe requirements. These functions are called **analytic functions**.

DEFINITION 3.2.2 Analyticity at a Point

A complex function $w = f(z)$ is said to be **analytic at a point** z_0 if f is differentiable at z_0 and at every point in some neighborhood of z_0.

A function f **is analytic in a domain** D if it is analytic at every point in D. The phrase "analytic *on* a domain D" is also used. Although we shall not use these terms in this text, a function f that is analytic throughout a domain D is called **holomorphic** or **regular**.

Very Important ▶ You should reread Definition 3.2.2 carefully. Analyticity at a point is *not* the same as differentiability at a point. Analyticity *at a point is a neighborhood property*; in other words, analyticity is a property that is defined over an open set. It is left as an exercise to show that the function $f(z) = |z|^2$ is differentiable at $z = 0$ but is not differentiable anywhere else. Even though $f(z) = |z|^2$ is differentiable at $z = 0$, it is not analytic at that point because there exists no neighborhood of $z = 0$ *throughout which* f is differentiable; hence the function $f(z) = |z|^2$ is nowhere analytic. See Problem 19 in Exercises 3.2.

In contrast, the simple polynomial $f(z) = z^2$ is differentiable at every point z in the complex plane. Hence, $f(z) = z^2$ is analytic everywhere.

Entire Functions A function that is analytic at every point z in the complex plane is said to be an **entire function**. In view of differentiation rules (2), (3), (7), and (5), we can conclude that polynomial functions are differentiable at every point z in the complex plane and rational functions are analytic throughout any domain D that contains no points at which the denominator is zero. The following theorem summarizes these results.

THEOREM 3.2.1 Polynomial and Rational Functions

(*i*) A complex polynomial function $p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$, where n is a nonnegative integer, is an entire function.

(*ii*) A complex rational function $f(z) = \dfrac{p(z)}{q(z)}$, where p and q are polynomial functions, is analytic in any domain D that contains no point z_0 for which $q(z_0) = 0$.

Singular Points Since the rational function $f(z) = 4z/\left(z^2 - 2z + 2\right)$ is undefined at $1 + i$ and $1 - i$, f fails to be analytic at these points. Thus by (*ii*) of Theorem 3.2.1, f is not analytic in any domain containing one or both of these points. In general, a point z at which a complex function $w = f(z)$ fails to be analytic is called a **singular point** of f. We will discuss singular points in greater depth in Chapter 6.

$1 \pm i$ are zeros of the denominator of f. ▶

Another consequence of the differentiation rules on page 123 is:

Analyticity of Sum, Product, and Quotient

If the functions f and g are analytic in a domain D, then the sum $f(z) + g(z)$, difference $f(z) - g(z)$, and product $f(z)g(z)$ are analytic in D. The quotient $f(z)/g(z)$ is analytic provided $g(z) \neq 0$ in D.

An Alternative Definition of $f'(z)$ Sometimes it is convenient to define the derivative of a function f using an alternative form of the difference quotient $\Delta w/\Delta z$. Since $\Delta z = z - z_0$, then $z = z_0 + \Delta z$, and so (1) can be written as

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}. \tag{12}$$

In contrast to what we did in Example 1, if we wish to compute f' at a general point z using (12), then we replace z_0 by the symbol z *after* the limit is computed. See Problems 7–10 in Exercises 3.2.

As in real analysis, if a complex function f is differentiable at a point, then the function is necessarily continuous at the point. We use the form of the derivative given in (12) to prove the last statement.

THEOREM 3.2.2 Differentiability Implies Continuity

If f is differentiable at a point z_0 in a domain D, then f is continuous at z_0.

Proof The limits $\lim_{z\to z_0} \dfrac{f(z)-f(z_0)}{z-z_0}$ and $\lim_{z\to z_0}(z-z_0)$ exist and equal $f'(z_0)$ and 0, respectively. Hence by Theorem 3.1.2(*iii*), we can write the following limit of a product as the product of the limits:

$$\begin{aligned}\lim_{z\to z_0}(f(z)-f(z_0)) &= \lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}\cdot(z-z_0)\\ &= \lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}\cdot\lim_{z\to z_0}(z-z_0) = f'(z_0)\cdot 0 = 0.\end{aligned}$$

From $\lim_{z\to z_0}(f(z)-f(z_0)) = 0$ we conclude that $\lim_{z\to z_0} f(z) = f(z_0)$. In view of Definition 3.1.2, f is continuous at z_0. ☰

Of course the converse of Theorem 3.2.2 is not true; continuity of a function f at a point does not guarantee that f is differentiable at the point. It follows from Theorem 3.1.3 that the simple function $f(z) = x + 4iy$ is continuous everywhere because the real and imaginary parts of f, $u(x, y) = x$ and $v(x, y) = 4y$ are continuous at any point (x, y). Yet we saw in Example 3 that $f(z) = x + 4iy$ is not differentiable at any point z.

As another consequence of differentiability, L'Hôpital's rule for computing limits of the indeterminate form $0/0$, carries over to complex analysis.

THEOREM 3.2.3 L'Hôpital's Rule

Suppose f and g are functions that are analytic at a point z_0 and $f(z_0) = 0$, $g(z_0) = 0$, but $g'(z_0) \neq 0$. Then

$$\lim_{z\to z_0}\frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}. \tag{13}$$

The task of establishing (13) is neither long nor difficult. You are guided through the steps of a proof in Problem 33 in Exercises 3.2.

EXAMPLE 4 Using L'Hôpital's Rule

Compute $\lim_{z\to 2+i}\dfrac{z^2-4z+5}{z^3-z-10i}$.

Solution If we identify $f(z) = z^2 - 4z + 5$ and $g(z) = z^3 - z - 10i$, you should verify that $f(2+i) = 0$ and $g(2+i) = 0$. The given limit has the indeterminate form $0/0$. Now since f and g are polynomial functions, both functions are necessarily analytic at $z_0 = 2 + i$. Using

$$f'(z) = 2z - 4, \quad g'(z) = 3z^2 - 1, \quad f'(2+i) = 2i, \quad g'(2+i) = 8 + 12i,$$

we see that (13) gives

$$\lim_{z\to 2+i}\frac{z^2-4z+5}{z^3-z-10i} = \frac{f'(2+i)}{g'(2+i)} = \frac{2i}{8+12i} = \frac{3}{26}+\frac{1}{13}i.$$ ☰

In part (b) of Example 4 in Section 3.1 we resorted to the lengthy procedure of factoring and cancellation to compute the limit

$$\lim_{z\to 1+\sqrt{3}i} \frac{z^2-2z+4}{z-1-\sqrt{3}i}. \tag{14}$$

A rereading of that example shows that the limit (14) has the indeterminate form $0/0$. With $f(z) = z^2 - 2z + 4$, $g(z) = z - 1 - \sqrt{3}i$, $f'(z) = 2z - 2$, and $g'(z) = 1$, L'Hôpital's rule (13) gives immediately

$$\lim_{z\to 1+\sqrt{3}i} \frac{z^2-2z+4}{z-1-\sqrt{3}i} = \frac{f'(1+\sqrt{3}i)}{1} = 2\left(1+\sqrt{3}i-1\right) = 2\sqrt{3}i.$$

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) In real calculus the derivative of a function $y = f(x)$ at a point x has many interpretations. For example, $f'(x)$ is the slope of the tangent line to the graph of f at the point $(x,\ f(x))$. When the slope is positive, negative, or zero, the function, in turn, is increasing, decreasing, and possibly has a maximum or minimum. Also, $f'(x)$ is the instantaneous rate of change of f at x. In a physical setting, this rate of change can be interpreted as velocity of a moving object. *None of these interpretations carry over to complex calculus*. Thus it is fair to ask: What does the derivative of a complex function $w = f(z)$ represent? Here is the answer: In complex analysis the primary concern is *not* what a derivative of function *is* or *represents*, but rather, it is whether a function f actually *has* a derivative. The fact that a complex function f possesses a derivative tells us a lot about the function. As we have just seen, when f is differentiable at z and at every point in some neighborhood of z, then f is analytic at the point z. You will see the importance of analytic functions in the remaining chapters of this book. For example, the derivative plays an important role in the theory of mappings by complex functions. Roughly, under a mapping defined by an analytic function f, the magnitude and sense of an angle between two curves that intersect a point z_0 in the z-plane is preserved in the w-plane at all points at which $f'(z) \neq 0$. See Chapter 7.

(*ii*) We pointed out in the foregoing discussion that $f(z) = |z|^2$ was differentiable only at the single point $z = 0$. In contrast, the real function $f(x) = |x|^2$ is differentiable everywhere. The real function $f(x) = x$ is differentiable everywhere, but the complex function $f(z) = x = \text{Re}(z)$ is nowhere differentiable.

(*iii*) The differentiation formulas (2)–(8) are important, but not nearly as important as in real analysis. In complex analysis we deal with functions such as $f(z) = 4x^2 - iy$ and $g(z) = xy + i(x + y)$, which, even if they possess derivatives, cannot be differentiated by formulas (2)–(8).

(*iv*) In this section we have not mentioned the concept of higher-order derivatives of complex functions. We will pursue this topic in depth in Section 5.5. There is nothing surprising about the definitions of higher derivatives; they are defined in exactly the same manner as in real analysis. For example, the second derivative is the derivative of the first derivative. In the case $f(z) = 4z^3$ we see that $f'(z) = 12z^2$ and so the second derivative is $f''(z) = 24z$. But there is a major difference between real and complex

variables concerning the existence of higher-order derivatives. In real analysis, if a function f possesses, say, a first derivative, there is no guarantee that f possesses any other higher derivatives. For example, on the interval $(-1,\ 1)$, $f(x) = x^{3/2}$ is differentiable at $x = 0$, but $f'(x) = \frac{3}{2}x^{1/2}$ is not differentiable at $x = 0$. In complex analysis, if a function f is *analytic* in a domain D then, by assumption, f possesses a derivative at each point in D. We will see in Section 5.5 that this fact alone guarantees that f possesses higher-order derivatives at all points in D. Indeed, an analytic function f is infinitely differentiable in D.

(*v*) The definition of "analytic at a point a" in real analysis differs from the usual definition of that concept in complex analysis (Definition 3.2.2). In real analysis, analyticity of a function is defined in terms of power series: A function $y = f(x)$ is analytic at a point a if f has a Taylor series at a that represents f in some neighborhood of a. In view of *Remark* (*iv*), why are these two definitions not really that different?

(*vi*) As in real calculus, it may be necessary to apply L'Hôpital's rule several times in succession to calculate a limit. In other words, if $f(z_0),\ g(z_0),\ f'(z_0)$, and $g'(z_0)$ are all zero, the limit $\lim\limits_{z\to z_0} f(z)/g\,(z)$ may still exist. In general, if f, g, and their first $n-1$ derivatives are zero at z_0 and $g^{(n)}(z_0) \neq 0$, then

$$\lim_{z\to z_0} \frac{f(z)}{g(z)} = \frac{f^{(n)}(z_0)}{g^{(n)}(z_0)}.$$

EXERCISES 3.2

Answers to selected odd-numbered problems begin on page ANS-11.

In Problems 1–6, use (1) of Definition 3.2.1 to find $f'(z)$ for the given function.

1. $f(z) = 9iz + 2 - 3i$

2. $f(z) = 15z^2 - 4z + 1 - 3i$

3. $f(z) = iz^3 - 7z^2$

4. $f(z) = \dfrac{1}{z}$

5. $f(z) = z - \dfrac{1}{z}$

6. $f(z) = -z^{-2}$

In Problems 7–10, use the alternative definition (12) to find $f'(z)$ for the given function.

7. $f(z) = 5z^2 - 10z + 8$

8. $f(z) = z^3$

9. $f(z) = z^4 - z^2$

10. $f(z) = \dfrac{1}{2iz}$

In Problems 11–18, use the rules of differentiation to find $f'(z)$ for the given function.

11. $f(z) = (2-i)z^5 + iz^4 - 3z^2 + i^6$

12. $f(z) = 5(iz)^3 - 10z^2 + 3 - 4i$

13. $f(z) = (z^6 - 1)(z^2 - z + 1 - 5i)$

14. $f(z) = (z^2 + 2z - 7i)^2(z^4 - 4iz)^3$

15. $f(z) = \dfrac{iz^2 - 2z}{3z + 1 - i}$

16. $f(z) = -5iz^2 + \dfrac{2+i}{z^2}$

17. $f(z) = \left(z^4 - 2iz^2 + z\right)^{10}$

18. $f(z) = \left(\dfrac{(4+2i)z}{(2-i)z^2 + 9i}\right)^3$

19. The function $f(z) = |z|^2$ is continuous at the origin.

(**a**) Show that f *is* differentiable at the origin.

(**b**) Show that f *is not* differentiable at any point $z \neq 0$.

20. Show that the function

$$f(z) = \begin{cases} 0, & z = 0 \\ \dfrac{x^3 - y^3}{x^2 + y^2} + i\dfrac{x^3 + y^3}{x^2 + y^2}, & z \neq 0 \end{cases}$$

is not differentiable at $z = 0$ by letting $\Delta z \to 0$ first along the x-axis and then along the line $y = x$.

In Problems 21 and 22, proceed as in Example 3 to show that the given function is nowhere differentiable.

21. $f(z) = \bar{z}$

22. $f(z) = |z|$

In Problems 23–26, use L'Hôpital's rule to compute the given limit.

23. $\lim\limits_{z \to i} \dfrac{z^7 + i}{z^{14} + 1}$

24. $\lim\limits_{z \to \sqrt{2}+\sqrt{2}i} \dfrac{z^4 + 16}{z^2 - 2\sqrt{2}z + 4}$

25. $\lim\limits_{z \to 1+i} \dfrac{z^5 + 4z}{z^2 - 2z + 2}$

26. $\lim\limits_{z \to \sqrt{2}i} z\dfrac{z^3 + 5z^2 + 2z + 10}{z^5 + 2z^3}$

In Problems 27–30, determine the points at which the given function is not analytic.

27. $f(z) = \dfrac{iz^2 - 2z}{3z + 1 - i}$

28. $f(z) = -5iz^2 + \dfrac{2 + i}{z^2}$

29. $f(z) = \left(z^4 - 2iz^2 + z\right)^{10}$

30. $f(z) = \left(\dfrac{(2 - i)z + 1}{z^2 + 9i}\right)^3$

Focus on Concepts

31. Use Definition 3.2.1 and Theorem 3.1.2 to prove the constant rule $\dfrac{d}{dz}cf(z) = cf'(z)$.

32. Use Definition 3.2.1 and Theorem 3.1.2 to prove the sum rule $\dfrac{d}{dz}[f(z) + g(z)] = f'(z) + g'(z)$.

33. In this problem you are guided through the start of the proof of the proposition:

If functions f and g are analytic at a point z_0 and $f(z_0) = 0,\ g(z_0) = 0$, but $g'(z_0) \neq 0$, then $\lim\limits_{z \to z_0} \dfrac{f(z)}{g(z)} = \dfrac{f'(z_0)}{g'(z_0)}$.

Proof We begin with the hypothesis that f and g are analytic at a point z_0. Analyticity at z_0 implies f and g are differentiable at z_0. Hence from (12) both limits,

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \quad \text{and} \quad g'(z_0) = \lim_{z \to z_0} \frac{g(z) - g(z_0)}{z - z_0}$$

exist. But since $f(z_0) = 0,\ g(z_0) = 0$, the foregoing limits are the same as

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z)}{z - z_0} \quad \text{and} \quad g'(z_0) = \lim_{z \to z_0} \frac{g(z)}{z - z_0}.$$

Now examine $\lim\limits_{z \to z_0} \dfrac{f(z)}{g(z)}$ and finish the proof.

34. In this problem you are guided through the start of the proof of the product rule.

Proof We begin with the hypothesis that f and g are differentiable at a point z; that is, each of the following limits exist:

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \quad \text{and} \quad g'(z) = \lim_{\Delta z \to 0} \frac{g(z + \Delta z) - g(z)}{\Delta z}$$